

PYTHON FOR AI TESTING

Fundamentals Assessment • Student Test

Student Name:	<u>Devika</u>
Date:	27 August 2026
Batch / ID:	_____
Duration:	90 Minutes
Maximum Marks:	100

INSTRUCTIONS

1. Attempt all questions. 2. Use meaningful variable names. 3. For coding questions, focus on correct syntax and appropriate data types. 4. AI-testing scenarios should show how Python fundamentals can prepare test data and validate results.

Coverage: variables, assignment, primitive and non-primitive data types, and common data-type methods, applied to AI testing.

SECTION A — MCQs

20 × 2 = 40 marks

1. Which statement correctly assigns the value 25 to age?

A. 25 = age B. age == 25 C. age = 25 D. int(age) = 25

2. What is the data type of True?

A. str B. bool C. int D. logical

3. Which is a non-primitive / collection data type?

A. int B. float C. list D. bool

4. What is x after x=10; x=20?

A. 10 B. 20 C. 1020 D. Error. // First, x gets 10, then, x = 20 reassigns x to 20.

5. Which string method converts text to lowercase?

A. lower() B. lowercase() C. down() D. small()

6. Which method removes whitespace from both ends of a string?

A. remove() B. strip() C. trim() D. clean()

7. Which type stores key-value pairs?

A. tuple B. set C. dictionary D. list

8. Which list method adds one item to the end?

A. add() B. append() C. insert_end() D. push()

9. Which creates a tuple?

A. [1,2,3] B. {1,2,3} C. (1,2,3) D. <1,2,3>

10. Which set method adds one element?

A. append() B. add() C. insert() D. push()

11. What does type(x) return?

A. value B. memory address C. type/class D. variable name

12. Which is immutable?

A. list B. dictionary C. set D. tuple

13. Which string method checks whether text starts with given text?

A. start() B. begins() C. startswith() D. prefix()

14. What does len([10,20,30]) return?

A. 2 B. 3 C. 30 D. 60 // len() returns the **number of elements** in a list.

15. Which dictionary method safely retrieves a missing key?

A. fetch() B. value() C. get() D. find()

16. For AI testing, why are lists useful?

A. store multiple test cases B. numbers only C. cannot change D. replace functions

17. What does text.replace('bad','good') do?

A. deletes text B. replaces matching text C. uppercase D. splits text

18. Which variable assignment is valid?

A. test-case=10 B. 10test=10 C. test_case=10 D. class=10

19. Which type suits a unique collection of labels?

A. set B. float C. bool D. string

20. In AI testing, boolean is especially useful for:

A. pass/fail B. long paragraph only C. records D. file path only

SECTION B — SHORT ANSWER & CONCEPTS

10 × 3 = 30 marks

21. Explain the difference between **variable assignment** and **comparison**. Give one example of each.

Ans: `x = 10`

10 is assigned to the variable x.

`x == 10`

Comparison is used to **check whether two values are equal**. It returns `True` or `False`

22. Name four primitive/basic Python data types and give one example value for each.

Ans: **int** — `age = 25`

float — `price = 10.5` **str** — `"Python"` **bool** — `is_passed = True` **complex** — `z = 2 + 3j`

23. Explain the difference between a **list** and a **tuple**. When would each be useful in test-data preparation?

Ans: A **list** is mutable, so its values can be changed. It is useful for test data that may need updates.

Ex: `list1=[10,20,30]`

A **tuple** is immutable, so its values cannot be changed. It is useful for fixed test data that should remain unchanged.

Ex: `tuple1=(10, 20, 30)`

24. Explain a **dictionary**. Create one AI test case with keys **input**, **expected**, and **category**.

Ans: A **dictionary** stores data in **key-value pairs**.

`test_case = { "input": "Hello", "expected": "Hi", "category": "Greeting" }`

25. What are **lower()**, **strip()**, and **replace()** used for? Give one testing use case for each.

Ans: **lower()** – Converts text to lowercase. **Testing use:** Compare AI responses without worrying about uppercase/lowercase.

strip() – Removes spaces from the beginning and end of text. **Testing use:** Clean user input before testing.

replace() – Replaces one text with another. **Testing use:** Modify or clean test data.

26. Explain **mutable vs immutable** with one Python example.

Ans: **Mutable:** A value that can be changed after it is created. **Example:** List `my_list = [1, 2, 3]` `my_list[0] = 10`

Immutable: A value that cannot be changed after it is created. **Example:** Tuple `my_tuple = (1, 2, 3)`

27. Why should an AI tester care about the **data type** of an input before sending it to an AI system?

A: An AI tester should care about the **data type** because the AI system may expect a specific type of input. Using the wrong data type can cause **errors or incorrect results**.

Example: A system expecting a number may not work correctly if it receives "10" as text instead of 10 as an integer.

28. Write at least five useful **list methods** and state what each does.

1. **append()** – Adds an item to the end. Example: `numbers = [1, 2]` → `numbers.append(3)` → `[1, 2, 3]`
2. **insert()** – Adds an item at a specific position. Example: `numbers = [1, 3]` → `numbers.insert(1, 2)` → `[1, 2, 3]`
3. **remove()** – Removes a specific item. Example: `numbers = [1, 2, 3]` → `numbers.remove(2)` → `[1, 3]`
4. **pop()** – Removes an item from the list. Example: `numbers = [1, 2, 3]` → `numbers.pop()` → `[1, 2]`
5. **sort()** – Sorts the list. Example: `numbers = [3, 1, 2]` → `numbers.sort()` → `[1, 2, 3]`

29. Write at least five useful **string methods** and state what each does.

1. **upper()** – Converts text to uppercase. Example: `"python".upper()` → `"PYTHON"`
2. **lower()** – Converts text to lowercase. Example: `"PYTHON".lower()` → `"python"`
3. **strip()** – Removes spaces from both ends. Example: `" Python ".strip()` → `"Python"`
4. **replace()** – Replaces text with other text. Example: `"I like Java".replace("Java", "Python")` → `"I like Python"`
5. **capitalize()** – Makes the first letter uppercase. Example: `"python".capitalize()` → `"Python"`

30. If an AI model should return PASS or FAIL, which Python data types could represent the result? Explain the advantages.

A **Boolean (bool)** is best for representing PASS or FAIL.

Example: `True = PASS`, `False = FAIL`.

Advantages: It is simple, clear, and easy to use in automated testing.

SECTION C — PYTHON CODING FOR AI TESTING

5 × 6 = 30 marks

28. Create variables for an AI test case: `prompt`, `expected_label`, `actual_label`, `temperature`, and `passed`. Use appropriate data types.

Answer / Code:

29. Normalize the following input by removing surrounding spaces, converting to lowercase, and replacing bad with good:

```
text = ' This is a BAD test case. '
```

Answer / Code:

30. Create a list containing five AI test prompts. Add one prompt, remove one prompt, and print the total number of prompts.

Answer / Code:

31. Create a dictionary containing `id`, `prompt`, `expected`, and `priority`. Then access and print `priority`.

Answer / Code:

32. Compare `expected` and `actual`. Store `True` in `passed` when they match, otherwise `False`. Print the result.

Answer / Code:

SECTION D — OPTIONAL AI TESTING CHALLENGE

Up to 10 bonus marks

33. An AI chatbot receives a string, integer, list, and dictionary as inputs. What Python checks would you perform before testing it?

34. You have 5 test cases stored as dictionaries inside a list. Why is this structure easier to maintain than five unrelated variables?

END OF TEST — Review your variable names, data types, and syntax.